Chapter 1 solutions

1

There are $26 - 2 = 24$ usable letters, chosen distinctly in $24 \cdot 23 \cdot 22$ ways. The first digit has nine choices. The second has nine remaining digits, hence

$$24 \cdot 23 \cdot 22 \cdot 9 \cdot 9 = 983\,664.$$

2

Choose the chair from eight, the vice chair from seven, then three members from the remaining six:

$$8 \cdot 7 \binom{6}{3} = 56 \cdot 20 = 1120.$$

Alternatively choose five people and assign the two ordered offices, giving $\binom{8}{5} \cdot 5 \cdot 4 = 1120$. Both descriptions retain ordered offices and unordered ordinary membership.

3

BANANARAMA has ten letters, including five A's, two N's, and one each of B, R, M. Fix A at each end. The middle eight positions contain three A's, two N's, and B, R, M, giving

$$\frac{8!}{3!2!} = 3360.$$

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4

The right side chooses a representative from n people, then an arbitrary subset of the other $n - 1$, giving $n2^{n-1}$. The left side first chooses a size-k subset, then its representative, and sums over $k = 0, \dots, n$. Each nonempty subset with a representative is counted exactly once by both methods, proving equality.

5

Put $y_i = x_i - 2 \geq 0$. Then

$$y_1 + y_2 + y_3 + y_4 = 5.$$

Stars and bars gives

$$\binom{5 + 4 - 1}{4 - 1} = \binom{8}{3} = 56.$$

6

$$C_6 = \frac{1}{7}\binom{12}{6} = \frac{924}{7} = 132.$$

Chapter 1 solutions (continued)

For the exact reflection, swap the prefix through the first height -1 of an inadmissible balanced word. The result has seven openings and five closings. Conversely, in any such word, swap through its first height $+1$, recovering an inadmissible balanced word. These maps are inverse, so $\binom{12}{5} = 792$ words are inadmissible, leaving $924 - 792 = 132$.

Division by two has no equal-pairing justification. The actual class sizes are 132 and 792. Swapping all parentheses sends a nonempty admissible word to an inadmissible one, but many inadmissible words remain inadmissible, such as `())(()` and `)(())(`. A pair need not contain exactly one admissible word.

7

Remove A's, leaving B, N, N, R, M with $5!/2! = 60$ orders. Each has six gaps; five A's choose distinct gaps:

$$60\binom{6}{5} = 360.$$

Deleting A's recovers the non-A order, and their original positions identify the selected gaps uniquely. Each resulting word is counted once.

With A at both ends, both exterior gaps are selected. Choose three of the four interior gaps:

$$60\binom{4}{3} = 240.$$

A's are indistinguishable, so do not multiply by ways to choose which two occupy the ends.

8

Four runs have two 1-runs and two 0-runs, with either starting symbol. Compositions of six ones into two positive lengths number $\binom{5}{1} = 5$, and those of three zeros number $\binom{2}{1} = 2$. Thus

$$2 \cdot 5 \cdot 2 = 20.$$

Five runs beginning with 1 have three 1-runs and two 0-runs, giving $\binom{5}{2}\binom{2}{1} = 20$. Beginning with 0 gives two 1-runs and three 0-runs, giving $\binom{5}{1}\binom{2}{2} = 5$. The disjoint cases total 25.

Without zeros, only 111111 exists, with one run. The formula assumes both symbols occur, so substituting $\binom{-1}{k}$ is invalid. If neither symbol occurs, the empty word has zero runs.

Chapter 1 solutions (continued)

9

For nonempty shelves, permute seven books and choose two cuts among six gaps:

$$7!\binom{6}{2} = 5040 \cdot 15 = 75\,600.$$

Concatenating shelves in label order uniquely recovers the order and cuts.

Allowing empty shelves, choose $a + b + c = 7$, $a, b, c \geq 0$, in $\binom{9}{2} = 36$ ways, then cut a permutation to those lengths:

$$7!\binom{9}{2} = 181\,440.$$

Recording destinations only gives $3^7 = 2187$. Erasing internal order has fiber size $a!b!c!$, which varies with shelf lengths. No fixed divisor converts the previous counts into this one.

10

Two of seven factors supply x, three supply $-2y$, and two supply z. The positional assignment count is

$$\frac{7!}{2!3!2!} = 210.$$

Each contributes weight $(-2)^3 = -8$, so the coefficient is $210(-8) = -1680$.

Set $x = y = z = 1$ to sum all coefficients, obtaining $(1 - 2 + 1)^7 = 0$. No set has negative size here. The positional choices have nonnegative counts; their signed weights are then added.

11

All words number $\binom{11}{4} = 330$. Reflecting the prefix through the first height -1 changes one excess negative step to positive. The target has eight positives and three negatives, giving $\binom{11}{3} = 165$.

Every target word ends at five, so it first reaches $+1$. Reflect that prefix to recover a word with seven positives and four negatives whose first negative height is -1. The operations are inverse. The admissible count is

$$330 - 165 = 165.$$

The answer happens to be half for these parameters. This does not justify general division by two; with seven positives and seven negatives, the admissible proportion is $1/8$.

Chapter 1 solutions (continued)

12

A full binary tree with five internal nodes has six leaves. Its ordered-subtree encoding $(U)V$ bijects it with five-pair admissible words, giving

$$C_5 = \frac{1}{6}\binom{10}{5} = 42.$$

For a heptagon, $n + 2 = 7$, hence $n = 5$ and $C_5 = 42$, through the root-triangle recurrence. Six fixed-order factors need five binary combinations, also giving 42.

Seven factors need six internal nodes and give

$$C_6 = \frac{1}{7}\binom{12}{6} = 132.$$

Parameters must follow the structures rather than merely matching numerals in the questions.

13

The map

$$(k, j, i) \longmapsto (k, j, i + 1)$$

gives $1 \le k < j < i + 1 \le n + 1$, a three-element subset of $\{1, \ldots, n + 1\}$. Subtracting one from the third coordinate inverts it, recovering $k < j \le i$. Thus the count is

$$\binom{n+1}{3}.$$

At $n = 8$, this is $\binom{9}{3} = 84$. The lecture allows $k = j$ as well, giving 120. Their difference, 36, is exactly the number of pairs $1 \le j \le i \le 8$, namely $\binom{9}{2}$. For $n < 2$, there are no triples, consistent with the out-of-range convention.

14

Set $y_1 = x_1 - 2, y_2 = x_2 - 1, y_3 = x_3, y_4 = x_4$. Their nonnegative total is $11 - 2 - 1 = 8$. Ignoring the upper bound gives

$$\binom{8 + 4 - 1}{4 - 1} = \binom{11}{3} = 165.$$

For inadmissible solutions $y_4 \ge 4$, put $z_4 = y_4 - 4 \ge 0$. Total four gives $\binom{7}{3} = 35$. The answer is

$$165 - 35 = 130.$$

Each inadmissible solution uniquely corresponds to a total-four solution. Only one violation class is subtracted, so overlapping complements are not an issue.

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Chapter 1 solutions (continued)

Verification code appendix: verify_counting.py

```
"""Recompute the small counting examples without the reference book."""

from itertools import permutations, product
from math import factorial


def catalan_enumerate(n: int) -> int:
    valid = 0
    for word in product("()", repeat=2 * n):
        balance = 0
        ok = True
        for ch in word:
            balance += 1 if ch == "(" else -1
            if balance < 0:
                ok = False
                break
        valid += int(ok and balance == 0)
    return valid


assert len({p for p in permutations("ABCDE", 3)}) == 5 * 4 * 3
assert 24 * 23 * 22 * 9 * 9 == 983_664
assert 8 * 7 * 20 == 1120
assert 6 * 5 * 84 + 9 * 15 * 28 == 6300
assert 15 * 84 * 5 == 6300
assert len(set(permutations("BANANARAMA"))) == factorial(10) // (
    factorial(5) * factorial(2)
)
assert factorial(8) // (factorial(3) * factorial(2)) == 3360
assert [catalan_enumerate(n) for n in range(7)] == [1, 1, 2, 5, 14, 42, 132]

print("counting examples: PASS")
```

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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https://github.com/xinhi6716/computer-science-self-study